

[core][distributed] add zmq fallback for broadcasting large objects #6183

New issue

Merged

youkaichao merged 11 commits into `vllm-project:main` from `youkaichao:add_zmq` on Jul 10, 2024

Conversation8

Commits11

Checks0

Files changed6

+274-80



youkaichao commented on Jul 7, 2024

Member

The input to vision language model contains images, which has variable length and can be quite large.

While the shared memory broadcast introduced in [#5399](#) works fine for LLMs, later we find we often need to adjust the buffer size for vision language models.

Estimating the size upper bound can be difficult. To solve the problem, this PR adds a fallback option using zeromq.

- When the object is small, we will use shared memory broadcast, which is fast and efficient.
- If the object is too large, we leave an overflow message in the shared memory, and send the data via zeromq, which can handle arbitrary sized data.

In addition, shared memory broadcast is limited to single node, while zeromq (socket-based) is not. Therefore, we can extend the broadcast to also work for cross-node settings. This PR extends the functionality.

cc [@DarkLight1337](#) [@ywang96](#) for vision language model related.

Reviewers

zhuohan123



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

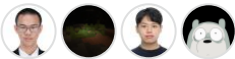
No milestone

Development

Successfully merging this pull request may close these issues.

None yet

4 participants



youkaichao added 9 commits 2 years ago

[update_in_the_same_node_as](#) [5da4157](#)

[add fallback to zmq for large data](#) [81e47ed](#)

[update tests](#) [61a4b82](#)

[bugfix](#) [dcf1e6d](#)

-   [bugfix](#) [b8251d5](#)
-   [bugfix](#) [0cf91d2](#)
-   [bugfix](#) [710c9a3](#)
-   [rename to message queue](#) [24e496d](#)
-   [update parallel state](#) [45d82fc](#)



youkaichao commented
on [Jul 7, 2024](#)

[View reviewed changes](#)

vllm/distributed/device_communicators/shm_broadcast.py

Comment on lines [+415](#) to [+425](#)

```
415 +         if self.n_local_reader > 0:
416 +             if len(serialized_obj) >
417 +                 with self.acquire_writer_lock():
418 +                     buf[0] = 1 # occupied
419 +                     self.local_socket.send(serialized_obj)
420 +             else:
421 +                 with self.acquire_writer_lock():
422 +                     buf[0] = 0 # not occupied
423 +                     buf[1:len(serialized_obj)] = serialized_obj
424 +             if self.n_remote_reader > 0:
425 +                 self.remote_socket.send(serialized_obj)
```



youkaichao on [Jul 7, 2024](#)

Member

Author

this is the most critical part, previously, when object size is too large (e.g. large image data), vLLM will error. Now, we will fall back to zmq.



DarkLight1337 commented on [Jul 8, 2024](#)

Member

• edited ▾

This idea sounds good, but I don't have much experience with cross-device communication, so I'll leave the review to someone who is more qualified.



zhuohan123 approved these
changes on [Jul 9, 2024](#)

[View reviewed changes](#)

zhuohan123 left a comment

Member

I took a very brief look and the code in general LGTM. I didn't check the internal logic of the message queue-based broadcast, but the interface and code change looks good. Please let me know whether I need to look into any part more carefully.

vllm/distributed/parallel_state.py

Outdated

Show resolved

 **youkaichao** added 2 commits 2 years ago

  rename daa49cc

  update comments 17d7e36



WoosukKwon commented on Jul 10, 2024

Collaborator

@youkaichao `pytest`
`tests/distributed/test_shm_broadcast.py` worked on the AMD MI210 machine. Do you want other kinds of tests as well?



youkaichao commented

Member Author

on Jul 10, 2024

@WoosukKwon thanks for testing! I think it works because I added the dependency in `requirements-common.txt` , so all platforms can use it.

  **youkaichao** merged commit **da78cae** into `vllm-project:main` on Jul 10, 2024

  **youkaichao** deleted the `add_zmq` branch 2 years ago

 **adityagoel14** pushed a commit to adityagoel14/vllm-torchrun-test that referenced this pull request on Jul 11, 2024

 [core][distributed] zmq fallback for broadcasting large objects (vllm.. f7b7547

...



 **ywang96** mentioned this pull request [on Jul 12, 2024](#)

**[Bug]: llava model gets stuck with
RuntimeError: Please increase the
max_chunk_bytes parameter. #6376**

🔒 Closed



xjpang pushed a commit to [xjpang/vllm](#) that referenced this pull request [on Jul 24, 2024](#)



[\[core\]\[distributed\] zmq fallback for
broadcasting large objects \(vllm...](#) [675d169](#) ...



mgoin mentioned this pull request [on Jul 26, 2024](#)

**[RFC]: Isolate OpenAI Server Into
Separate Process #6797**

🔒 Closed



Alvant pushed a commit to [compressa-ai/vllm](#) that referenced this pull request [on Oct 26, 2024](#)



[\[core\]\[distributed\] zmq fallback for
broadcasting large objects \(vllm...](#) [e402263](#) ...



LeiWang1999 pushed a commit to [LeiWang1999/vllm-bitblas](#) that referenced this pull request [on Mar 26, 2025](#)



[\[core\]\[distributed\] zmq fallback for
broadcasting large objects \(vllm...](#) [9a3bd83](#) ...